

Muhammad Saram Hassan

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RESEARCH INTERESTS

LLM and Agent Security, Prompt Injection and Adversarial Attacks, Web Privacy and Transparency, Multi-Agent Systems, Trustworthy Evaluation, AI Safety & Policy.

EDUCATION

Lahore University of Management Sciences (LUMS)

Aug 2022 - May 2026

B.S. Computer Science, *Graduated with Distinction*

Lahore, Pakistan

- **Dean's Honor List (2022-2026).**
- **Graduate-level coursework:** Topics in Network Security, Advanced Topics in ML, Distributed Systems, Deep Learning, Generative AI.

PUBLICATIONS & MANUSCRIPTS

Peer-Reviewed

Understanding or Imitation? Auditing Conceptual Understanding and Reasoning in Large Language Models

M. S. Hassan, E. Jan, R. Kaur, E. Yeh, F. Zaffar, A. Gehani.

ACM REP 2026 [Paper] [Code]

- Reproduced and extended “Potemkin Understanding in LLMs” (ICML 2025) claiming models can define a concept correctly and still fail to use it across 32 concepts, 3,159 annotated instances, and 4 task types reconstructing the inference pipeline from scratch.
- Identified **four sources of methodological fragility**: run-to-run variance up to 51% (enough to flip above/below-chance conclusions), an unbounded metric, a silent $\sim 3B$ -parameter capability floor, and self-judging circularity that makes reported rates a lower bound on a lower bound.
- Observed that reasoning models significantly reduce this incoherence suggesting that these failures are driven by the inference regime rather than the underlying model itself.

Data Doping or True Intelligence? Evaluating the Transferability of Injected Knowledge in LLMs

E. Jan[†], M. Ali[†], M. S. Hassan, F. Zaffar, Y. Zaki.

Findings of EMNLP 2025 [Paper] [arXiv]

- Causal study of how fine-tuning *task format* governs knowledge internalization with factual content held fixed. Proved that comprehension-intensive tasks (QA 48%) far exceed token-mapping tasks (translation 17%) across 5 model families with monotonic scaling from 1.5B to 72B.
- Observed that retained facts are not generalized e.g models which could answer direct questions about a fact dropped to about 24% when asked questions that required using the fact in a new context. Thus, passing a recall test is not evidence that the model actually learned anything usable.

Under Submission

Decomposing Error in Continual World Models

M. S. Hassan, M. Irfan, F. Zaffar.

Under review, *Continual World Models Workshop, NeurIPS 2026*.

- When the world changes and a model is updated to keep up, we currently check only whether the updated model is accurate. That does not tell us whether the update tracked the actual change. A model can score well by throwing away everything it knew and refitting on new data, quietly “fixing” things that never broke. The Endpoint error could thus be decomposed as $\varepsilon^1 = \varepsilon^0 + (\delta_M - \delta_E)$: inherited error plus *update mismatch* on a predeclared interventional query question set. Neither number is enough on its own. Chasing mismatch alone keeps every old mistake forever, and accuracy alone cannot tell a careful revision from a full rebuild.
- In a controlled linear system, we show the two disagree completely and a small neural network on a simulated pendulum reproduces the effect. We propose reporting both, plus the pre-existing error, as a protocol.

In Preparation

SoK: A Systematization of Prompt Injection Threats in LLM-Powered Agents

M. S. Hassan, A. Usman, T. Khan, F. Zaffar.

Target venue IEEE S&P 2027.

- Unifying threat model over **84 works** (13 quantitative benchmarks), with a **two-axis, 12-class attack taxonomy** (channel $\times$ payload construction) and a four-pillar defense taxonomy evaluated per attack class.
- **Structural argument**: agentic systems collapse user instructions, tool outputs, retrieved documents, memory, and rendered UI into one token sequence with no control/data separation, so defenses must carry *authority and provenance* metadata rather than read content better.
- **Central finding is the “reasoning gap”**. capability improves robustness to direct injection ($<2\%$ ASR) but not to context or perception corruption (55.1% memory-injection ASR; dark-pattern susceptibility rises $37.6\% \rightarrow 71.2\%$ when reasoning budget increases alone).

- Built a synthetic adversarial HTML testbed (MCP poisoning, DOM-obfuscated injection, visual spoofing). We also found that LLM browsers bypass robots.txt while masquerading as human users and cannot be behaviorally differentiated using current fingerprinting protocols.

Escalate, Don't Improvise: Comparability in Autonomous Measurement Fleets

M. S. Hassan M. Pervez, Y. Zaki.

Full draft complete; target venue **HotMobile**.

- Built **AmiClaw**: a volunteer-phone Internet-measurement platform where each node runs an on-device LLM agent against a *specification* rather than a fixed script bounded by a model-free reference monitor (that tracks allowlists, rate/budget caps, declared parameter ranges) so limits are structural and not prompt-dependent.

SENTRY: Adversarial Agent Detection in Open Multi-Agent Systems

M. S. Hassan, A. Gehani. target venue *ICLR*.

- Real multi-agent systems have mixed-observability. A few agents you run yourself and can inspect internally with many commercial APIs you can only send text to. If one agent sabotages the group, existing defenses either need to inspect every agent, or read only text and miss anything subtle. We reframe the problem to now stop probing agents you cannot open and treat white-box agents as sensors and record how their internal states react when they read a message from a black-box one.

RESEARCH EXPERIENCE

Research Assistant - SRI International

Aug 2025 – Present

Advisors: Dr. Ashish Gehani, Dr. Ramneet Kaur, Dr. Eric Yeh

Remote / Menlo Park, CA

- Set up and ran the lab's GPU infrastructure: environment and driver setup, model sharding across cards, batching and memory tuning to fit varying parameter models, and storage for the internal activations these experiments generate.
- Built the paper's evaluation pipeline from scratch, independently selecting models, identifying key design questions, and designing experiments to resolve them.

Research Assistant - Internet Security & Privacy Lab, LUMS

Jan 2025 – Feb 2026

Advisors: Dr. Fareed Zaffar (LUMS), Dr. Taha Khan (CMU CyLab)

Lahore, Pakistan

- Authored the SoK (above), synthesizing attack success rates across 12+ agent-security benchmarks and built the adversarial HTML testbed.
- Showed these benchmark numbers are not comparable across studies despite being cited side by side by extracting for each what the attacker manipulates and how compromise is measured.

Research Intern - CommNets Lab (NYU Abu Dhabi)

Jan 2025 – Present

Advisors: Dr. Yasir Zaki (NYUAD)

Remote/ Abu Dhabi, UAE

- Designed and deployed **AmiClaw** on a five-node testbed spread across two regions, where every node is an ordinary mobile phone on a real carrier network rather than a server. Handled the end-to-end setup: on-device agent runtime, measurement tooling per device, network permissions, and result collection back to a central store.
- Configured the fleet to run from a single point of instruction: one researcher-approved campaign plan and one guardrail policy go out to every node, each node resolves that plan against its own hardware and network, and anything it is not authorised to decide comes back to the orchestrator, which answers once and propagates that answer to every other node. This keeps the fleet from drifting apart one device at a time.
- Earlier: built the dataset and fine-tuning pipeline behind our EMNLP paper: 126 verified facts, a second crowd-sourced test set designed so memorising the training phrasing does not help, and training and evaluation runs across five model families and four task formats.

Research Intern - CITY Lab, LUMS

Sep 2024 – May 2025

Advisor: Dr. Muhammad Tahir / Interpretable, property-aware neural-network pruning

Lahore, Pakistan

- Built a **property-aware pruning** framework (VGG-19 / CIFAR-100) using transformed-input importance metrics and a filter-locking mechanism that protects OOD-relevant filters: 62.47% vs. 60.64% baseline at 80% compression, with reduced deep-layer representational drift.

TEACHING

- **Teaching Assistant**, CS-210 Discrete Mathematics (Dr. Malik Jahan Khan) & CS-100 Computational Problem Solving (Dr. Fareed Zaffar); held weekly office hours for 300+ students.
- **Instructor**, Fundamentals of Machine Learning (LUMS Science School & Uplift AI) and Computational Space Science (FAST University); designed original curricula, 100+ students each.

HONORS & AWARDS

- **Dean's Honor List**, LUMS; **100% Merit Scholarship** (A-Levels).
- **Gold Medal**, International Kangaroo Linguistics Contest 2021 (3000+ competitors); **Silver Medal**, International Youth Mathematical Championship 2020 (12,500+ participants, 98 countries); **Top-30 with Honorable Mention** (only Pakistani team) at CERN Beamline for Schools 2021.

TECHNICAL SKILLS

Languages: Python, C/C++, Go, Java, SQL, Bash, C#, JavaScript. **ML/Systems:** PyTorch, TensorFlow, HuggingFace, vLLM, LoRA/PEFT, multi-GPU training and inference, distributed systems, Docker, Linux.